

NutriScan AI

Designing an instant AI meal and nutrition scanner

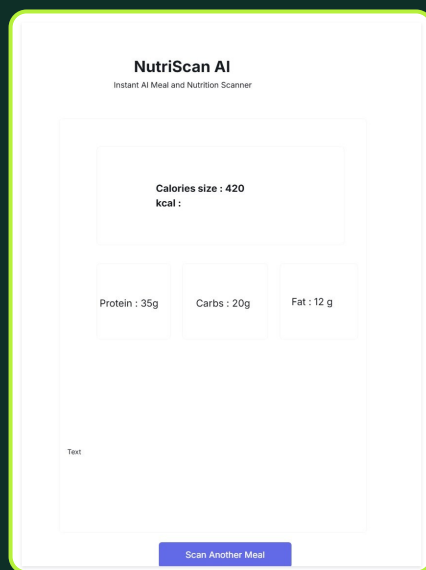
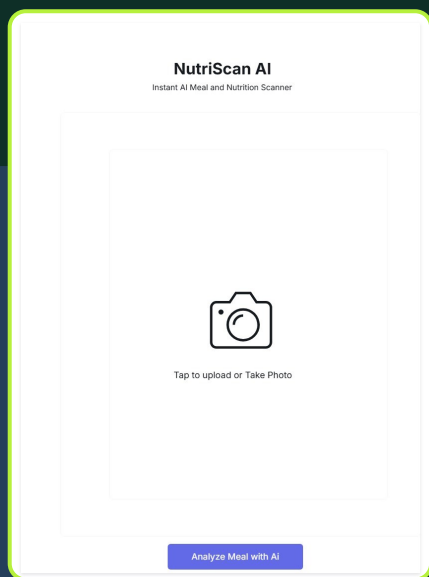
A 5-screen mobile concept: snap a photo of a meal, get calories, macros, ingredients and allergens, and keep a daily log. Wireframed in Visily, with a product review of what to refine before building.

Visily

UX flow

AI product thinking

Wireframe



Salma Mazhar

UX concepts | AI product thinking | Education and wellness

Project Overview

The problem, the idea and who it is for

Tracking food is tedious. People skip it because typing every ingredient and guessing portions takes too long. NutriScan AI explores a simple alternative: take one photo and let AI do the first pass.

The problem

Manual food logging is slow, so people quit within days and never see how their meals add up.

The idea

One photo gives calories, protein, carbs, fat, ingredients and allergens, then saves the meal to a daily log.

Who it is for (assumed users)

Busy professionals

Want a quick check on meals without opening a food database.

Fitness beginners

Need simple calorie and protein feedback while building habits.

People with food allergies

Want an extra check on what a meal might contain.

Parents and students

Plan balanced meals without complicated tools.

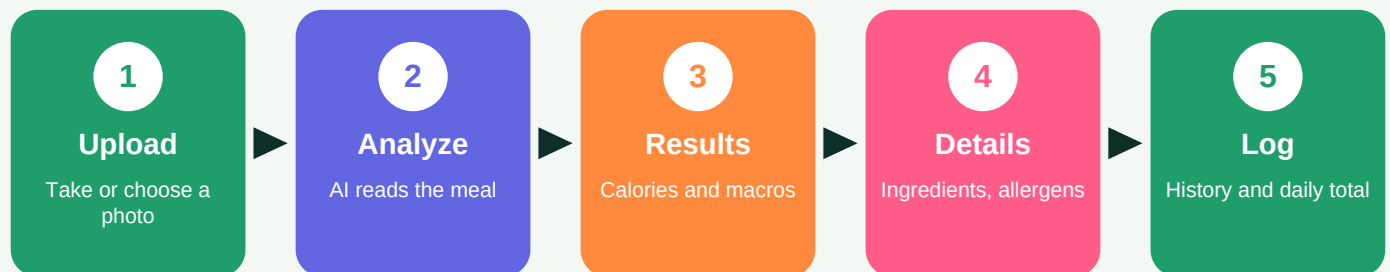
Project scope

This is a concept and wireframe, not a working app. These users are my assumptions, not interview results. No real AI model or accuracy testing was done, and this case study does not claim any.

The User Flow

Five screens from photo to daily log

I kept the journey short: one primary action per screen, with a single button that always moves the user forward.



Button labels follow the user's next step

Analyze Meal with AI

Screen 1 to 2

Names the action and the AI's role.

Scan Another Meal

Screen 3

Lets users repeat without going back.

View Meal History

Screen 4

Moves from one meal to the bigger picture.

Log New Meal

Screen 5

Closes the loop and starts the next scan.

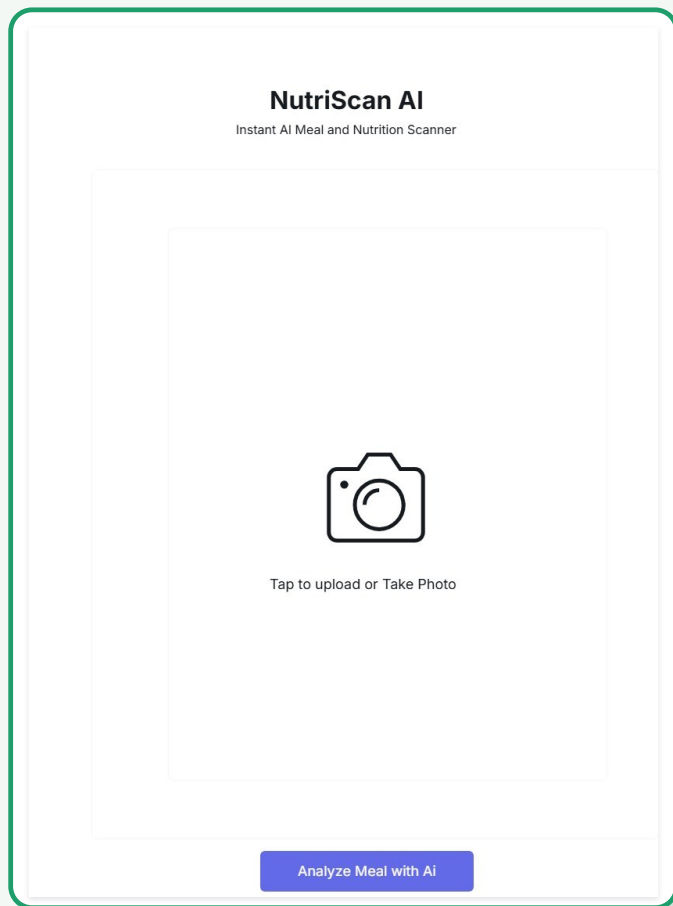
Design principle

One screen, one job. The camera screen only asks for a photo; the results screen only shows numbers; details live on their own screen so nothing feels crowded.

Screens 1 and 2: Capture

Making the first step effortless

The home screen is almost empty on purpose. A large camera icon and the line 'Tap to upload or Take Photo' make the first action obvious. A loading screen then tells the user the AI is working.



Screen 1: upload or take a photo.



Screen 2: analyzing state with a spinner.

Why it works

A single, large tap target and plain wording suit a first-time user. The primary button 'Analyze Meal with AI' is in the same place on every screen.

Why a loading state matters

AI takes a moment. Showing progress tells users the app has not frozen and builds trust before the result appears.

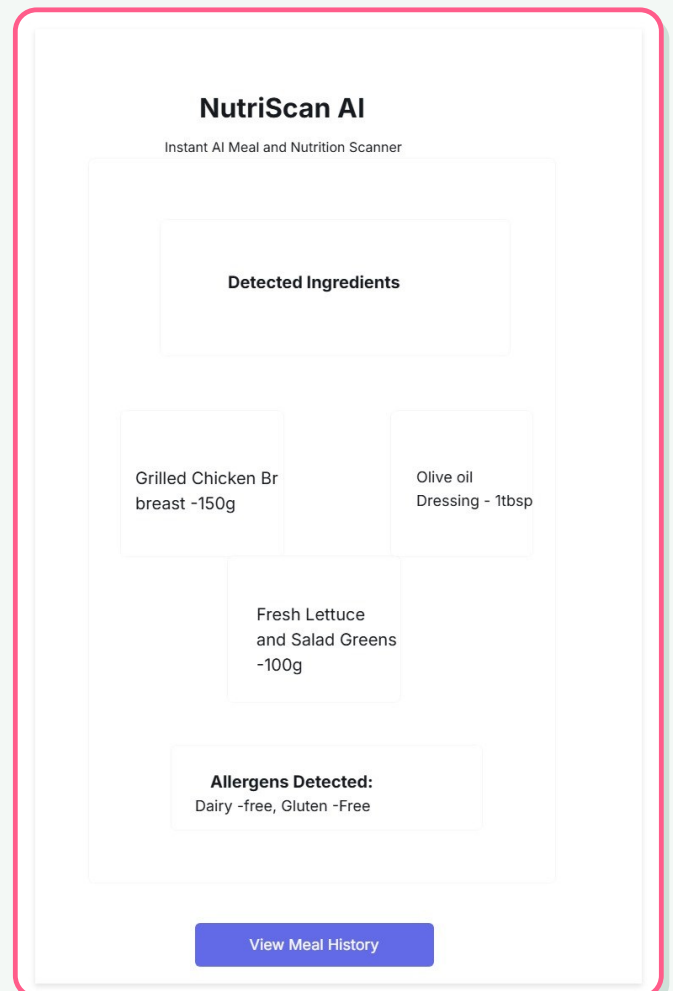
Screens 3 and 4: Results

Numbers first, then the explanation

The results screen puts calories at the top, with protein, carbs and fat below as three equal tiles. The next screen explains how the AI got there by listing ingredients with portions and flagging allergens.



Screen 3: calories and macros.



Screen 4: detected ingredients and allergens.

Layered information

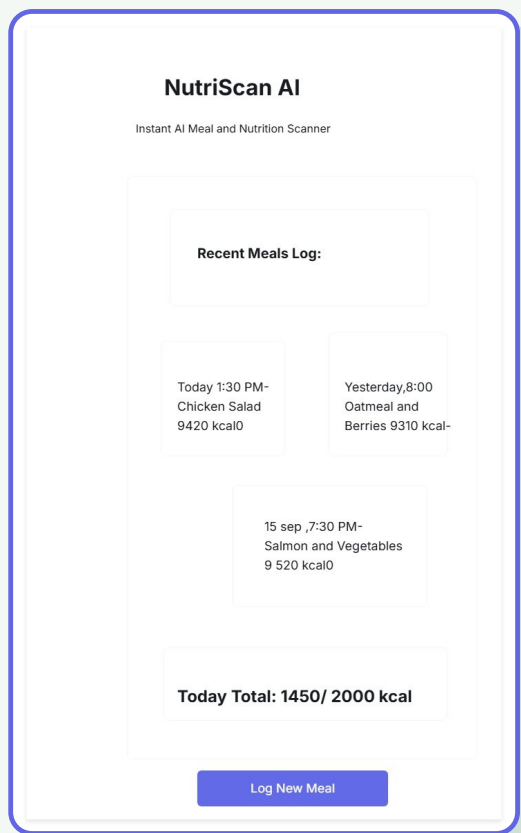
Most people only need the headline number. Ingredients and allergens sit one tap deeper, so the main view stays clean.

Transparency builds trust

Showing detected ingredients with weights lets users see how the AI reached the calorie number and spot mistakes.

Screen 5 and the AI Logic

The daily log and the product logic behind it



Screen 5: meal history and daily total.

The log lists recent meals with time, name and calories, and ends with a daily total against a goal ('1450 / 2000 kcal'). Seeing progress toward a goal is what turns a one-time scan into a habit.

Proposed AI pipeline (not built)

Photo

Vision model finds foods

Estimate portion sizes

Match to nutrition data

Show results + ask to confirm

What the product manager must watch

Accuracy

Photos cannot show hidden oil, sauces or exact portion size, so numbers are estimates.

Allergen safety

A wrong 'allergen-free' label can cause real harm. Show it as a guide and add a clear disclaimer.

User control

Let users edit ingredients and portions so the AI's first guess can be corrected.

Privacy

Meal photos and health habits are personal data and need clear consent.

Design Review: What to Fix

An honest look at the wireframe

A review is part of the work. Reading each screen closely shows details to polish before a developer or client sees it.

Text errors on screens

High

'Circulating Nutrution' (likely 'Calculating Nutrition'), 'Calories size : 420 kcal :' and a stray 'Text' placeholder on the results screen.

Calorie numbers look wrong

High

The log shows '9420 kcal0', '9310 kcal-' and '9 520 kcal0' (probably 420, 310 and 520). Today's total says 1450 although only one meal is listed for today.

Layout is uneven

Med

Ingredient and meal cards sit at different heights, and 'Grilled Chicken Br breast' is cut off. A consistent grid would look more professional.

No way to correct the AI

Med

There is no edit button for ingredients or portions, and no confidence indicator for the estimate.

Missing safety note

Med

Allergen results appear without a disclaimer that they are estimates. Wording like 'Dairy-free' should read 'No dairy detected'.

Missing states

Low

No error screen for a blurry photo or failed scan, and no empty state for a new user with no meals yet.

Roadmap, Metrics and Lessons

From wireframe to a testable product

Next steps

1

Fix text, numbers and alignment in the wireframe

Quick clean-up before sharing.

2

Add edit, confidence and error screens

Give users control over the AI's guess.

3

Interview 5 to 8 target users

Replace assumptions with evidence.

4

Build a clickable prototype and test with real meal photos

Check whether estimates are useful.

How I would measure success (proposed)

Scan-to-save rate

Share of scans that end with a saved meal.

Edit rate

How often users correct the AI. A high rate means weak accuracy.

7-day return

Users who come back to log within a week.

What I learned

A good AI product starts with a simple user flow, not the model. Trust comes from showing how the AI decided and letting users correct it. And reviewing your own design, line by line, catches problems early.

Salma Mazhar

Message me to discuss an AI product or UX project.